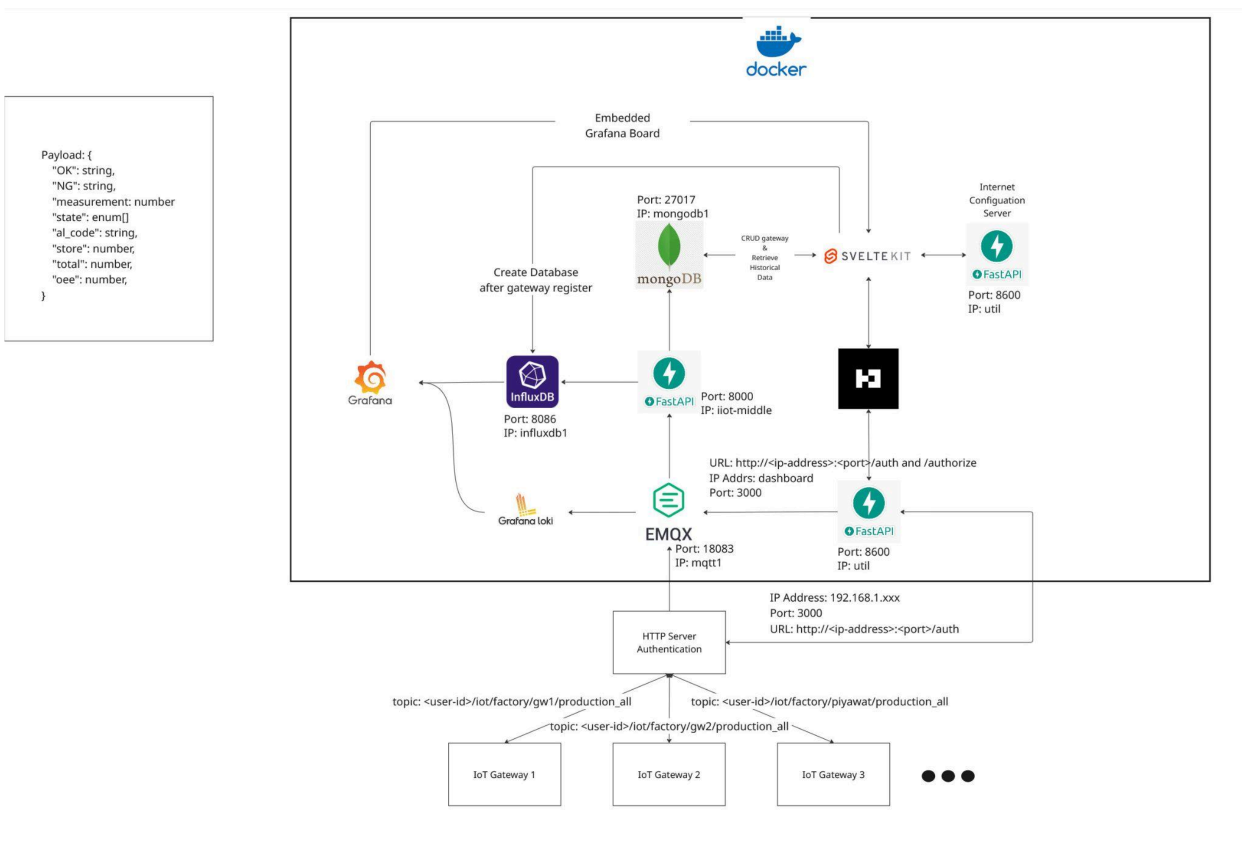
IIoT Gateway Connection and Configuration Guide

Overview

This document outlines the technical specifications required to connect an Industrial Internet of Things (IIoT) gateway to the central data connector server. It details the network configuration, authentication protocols, payload schemas, and visualization access credentials necessary for system testing and evaluation.



1. Network Connectivity Specifications

To establish a connection to the MQTT broker and the data connector application, configure the client with the following settings.

Parameter	Value	Description
MQTT Broker Host	192.168.1.137	The IP address of the local server.
MQTT Broker Port	1883	Standard MQTT TCP port.
Security Protocol	Username / Password	Authentication is strictly credential-based.
Fallback Credentials	admin@gmail.com / admin123	System administrator credentials used if gateway-specific authentication fails.

2. Gateway Authentication and Topic Routing

The system utilizes strict topic-based routing. Devices must publish to the specific topic associated with their credentials to ensure data is processed correctly. The topic strings include a unique identifier (UUID) required by the backend.

Device Name	Username / Password	Target MQTT Topic (Publish)
Piyawat Gateway	piyawat@gmail.com password123	691c1f4c2b2333b7705b88f8/iot/factory/piyawat/production_all
Gateway 1	gateway1@gmail.com password123	691c1f162b2333b7705b88f2/iot/factory/gw1/production_all
Gateway 2	gateway2@gmail.com password123	691c1f252b2333b7705b88f4/iot/factory/gw2/production_all

3. Payload Data Structure

The data connector expects the MQTT message payload to be a single JSON object. The system supports key normalization to handle variable naming conventions from different sensors.

A. JSON Schema

The payload must contain a dictionary where the keys correspond to the monitored production lines: production1 or production2.

Example Valid Payload:

JSON


```
{
  "Data": {
    "production1": {
      "Total": 100,
      "OK": 95,
      "Ng": 5,
      "OEE": 85.0,
      "State": "1"
    },
    "production2": {
      "P": 88.0,
      "A": 92.0,
      "Quality": 95.0,
      "state": "1"
    }
  }
}
```

B. Field Normalization Dictionary

The system automatically normalizes incoming data keys to a standard format for database storage. You may use any of the accepted aliases listed below.

Standardized Key	Accepted Aliases (Case-Insensitive)	Description
ok	ok, ok1, good, pass, passed, success	Count of successful products.
ng	ng, ng1, bad, fail, failed	Count of non-compliant products.
quality	quality, Q, %Q	Quality percentage or value.
performance	performance, P, %P	Performance percentage or value.
availability	availability, A, %A	Availability percentage or value.
oee	oee, OEE, %OEE	Overall Equipment Effectiveness value.
total	total	Total production count.
state	state	Current operational state (e.g., RUNNING, IDLE).
alarmcode	alarmcode	Active machine alarm or error codes.

4. Error Handling

The system includes an automated error reporting mechanism for invalid data ingestion.

1. **Invalid Payloads:** Data that is not valid JSON, uses an incorrect topic, or contains invalid line keys is rejected.
2. **Logging:** Rejected payloads are stored in the invalid_payloads MongoDB collection for audit purposes.
3. **Notification:** An error notification is published to the following MQTT topic:
    - iot/factory/errors

5. System Visualization Access (Grafana)

These credentials provide access to the Grafana web interface for dashboard visualization and data analysis.

Access URL: http://192.168.1.137:3000 (Default configuration)

Role	**Username**	**Password**	**Permissions**
Evaluator / Admin	professor	234Cy&<G7Mwh0y6J	**Full Viewer** (Access to all system dashboards).
Operator 1	gateway1	gateway1	Restricted to Gateway 1 specific data.
Operator 2	gateway2	gateway2	Restricted to Gateway 2 specific data.

Appendix A: Manual Verification Commands (CLI)

To validate the system without using physical sensors, the following mosquitto_pub commands can be executed from a terminal. These commands simulate various data scenarios, including standard production, key normalization, and error handling.

Note: Ensure you replace <HOST_IP> with your actual server IP (e.g., 192.168.1.137) and use the correct credentials defined in Section 2.

Test Case 1: Standard Data Ingestion (Gateway 2)

Action: Publishes a complete, standard payload. *Expected Result:* Data appears on the Gateway 2 Dashboard with "Running" state.

```
mosquitto_pub -h 192.168.1.137 -p 1883 -u "gateway2@gmail.com" -P "password123" \
-t "691c1f252b2333b7705b88f4/iot/factory/gw2/production_all" -m '{
  "Data": {
    "production1": {
      "total": 100,
      "Good": 500,
      "ng": 10,
      "store": 14,
      "quality": 100,
      "performance": 95,
      "availability": 98,
      "oee": 100
    }
  }
}'
```

Test Case 2: Schema Normalization (Prof. Piyawat Gateway)

Action: Publishes data using aliases (Q, %P, Good) instead of standard keys. *Expected Result:* System accepts the data and maps Q → quality, %P → performance, and Good → ok.

```
mosquitto_pub -h 192.168.1.137 -p 1883 -u "piyawat@gmail.com" -P "password123" \
-t "691c1f4c2b2333b7705b88f8/iot/factory/piyawat/production_all" -m '{
  "Data": {
    "production2": {
      "total": 300,
      "Good": 950,
      "ng": 40,
      "state": 1,
      "alarmcode": 0,
      "Q": 95.0,
      "%P": 97.0,
      "Availability": 99.0
    }
  }
}'
```

Test Case 3: Partial Data Update

Action: Publishes a payload with missing fields (only Quality is sent). *Expected Result:* The dashboard updates the Quality metric, while other metrics retain their previous state or show as null/zero depending on dashboard configuration.

```
mosquitto_pub -h 192.168.1.137 -p 1883 -u "gateway2@gmail.com" -P "password123" \
-t "691c1f252b2333b7705b88f4/iot/factory/gw2/production_all" -m '{
  "Data": {
    "production1": {
      "quality": 90
    }
  }
```

```
}'
```

Test Case 4: Negative Testing (Invalid Topic)

Action: Publishes data to an unknown or unregistered topic path. *Expected Result:* The data is **not** processed, and an entry appears in the system error log or the iot/factory/errors topic.

```
mosquitto_pub -h 192.168.1.137 -p 1883 -u "gateway2@gmail.com" -P "password123" \
-t "iot/factory/unknown/production_all" -m '{
  "Data": {
    "production1": {
      "quality": 100
    }
  }
}'
```

Test Case 5: Negative Testing (Invalid Line Key)

Action: Publishes data with an incorrect production line key (productionX). *Expected Result:* The payload is rejected and logged to the invalid_payloads collection in the database.

```
mosquitto_pub -h 192.168.1.137 -p 1883 -u "gateway2@gmail.com" -P "password123" \
-t "691c1f252b2333b7705b88f4/iot/factory/gw2/production_all" -m '{
  "Data": {
    "productionX": {
      "total": 100,
      "quality": 90
    }
  }
```

```
}'
```

Working payload:

Payload 1:

```
mosquitto_pub -h 192.168.1.137 \
  -t "691c1f252b2333b7705b88f4/iot/factory/gw2/production_all" \
  -u "gateway2@gmail.com" -P "password123" \
   -m  '{"Data":  {"production2":  {"quality":  91,  "performance":  87,
"availability": 94}}}'
```

Payload 2:

```
mosquitto_pub  -h  192.168.1.137  -p  1883    -u  "piyawat@gmail.com"  -P
"password123"                                                            -t
"691c1f4c2b2333b7705b88f8/iot/factory/piyawat/production_all"   -m '{
    "Data": {
      "production1": {
        "ok": 200,
        "ng": 10,
        "total": 210,
        "state": 0,
        "alarmcode": 1,
        "measurement": 80,
        "store": 18,
        "quality": 92,
        "performance": 88,
        "availability": 95
      },
      "production2": {
        "ok": 190,
        "ng": 10,
        "total": 200,
        "state": 0,
        "alarmcode": 1,
        "measurement": 75,
```

```
        "store": 15,
        "quality": 92,
        "performance": 88,
        "availability": 97
      }
    }
  }'
```

Payload 3:

```
mosquitto_pub -h 192.168.1.137 -p 1883 \
-u "piyawat@gmail.com" -P "password123" \
-t  "691c1f4c2b2333b7705b88f8/iot/factory/piyawat/production_all"  -m
'{
  "Data": {
    "production1": {
      "total": 210,
      "ok": 200,
      "ng": 10,
      "state": 0,
      "alarmcode": 1,
      "store": 18,
      "measurement": 80,
      "quality": 92,
      "performance": 88,
      "availability": 95,
      "oee": 90
    }
  }
}'
```

Payload 4:

```
mosquitto_pub -h 192.168.1.137 -p 1883 \
-u "piyawat@gmail.com" -P "password123" \
```

```
-t  "691c1f4c2b2333b7705b88f8/iot/factory/piyawat/production_all"  -m
'{
  "Data": {
    "production1": {
      "quality": 96
    }
  }
}'
```

Payload 5:

```
mosquitto_pub -h 192.168.1.137 \
-t "691c1f4c2b2333b7705b88f8/iot/factory/piyawat/production_all" \
-u "piyawat@gmail.com" -P "password123" \
-m '{
  "Data": {
    "production1": {
      "total": 300,
      "ok": 290,
      "ng": 10,
      "quality": 95,
      "performance": 92,
      "availability": 97,
      "oee": 94,
      "state": 1,
      "alarmcode": 0,
      "store": 25
    }
  }
}'
```

Payload 6:

```
mosquitto_pub -h 192.168.1.137 \
-t "691c1f4c2b2333b7705b88f8/iot/factory/piyawat/production_all" \
-u "piyawat@gmail.com" -P "password123" \
```

```
-m '{
  "Data": {
    "production1": {
      "quality": 100,
      "performance": 85,
      "availability": 90
    }
  }
}'
```